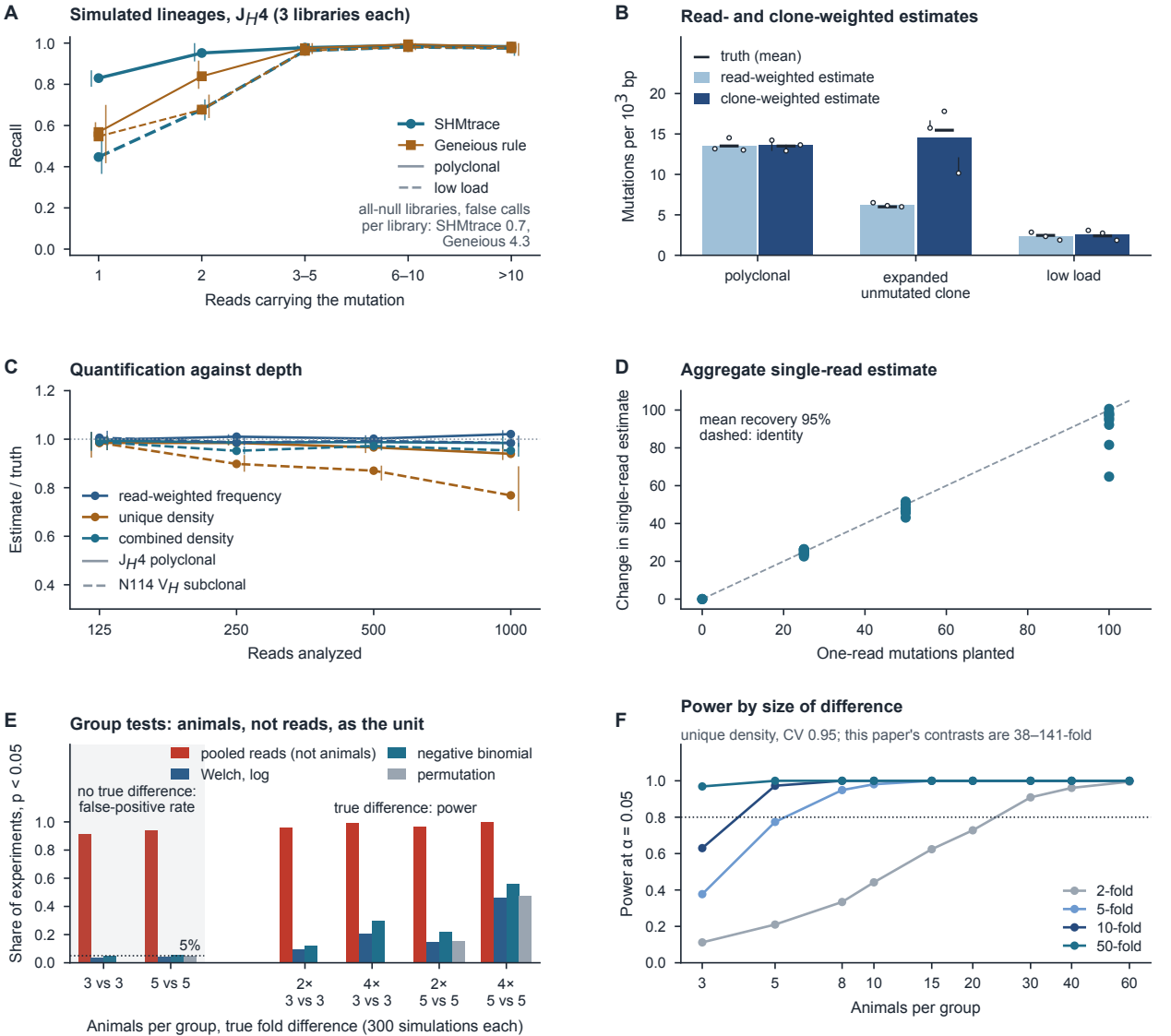

A
Simulated lineages, J_H4 (3 libraries each)
Recall
1.0
0.8
0.6
0.4
0.2
0.0
SHMtrace
Geneious rule
polyclonal
low load
all-null libraries, false calls
per library: SHMtrace 0.7,
Geneious 4.3
1
2
3–5
6–10
>10
Reads carrying the mutation
B
Read- and clone-weighted estimates
truth (mean)
read-weighted estimate
clone-weighted estimate
Mutations per 10^3 bp
20
15
10
5
0
polyclonal
expanded
unmutated clone
low load
C
Quantification against depth
Estimate / truth
1.2
1.0
0.8
0.6
0.4
read-weighted frequency
unique density
combined density
J_H4 polyclonal
N114 V_H subclonal
125
250
500
1000
Reads analyzed
D
Aggregate single-read estimate
mean recovery 95%
dashed: identity
Change in single-read estimate
100
80
60
40
20
0
0
20
40
60
80
100
One-read mutations planted
E
Group tests: animals, not reads, as the unit
pooled reads (not animals)
Welch, log
negative binomial
permutation
no true difference:
false-positive rate
true difference: power
Share of experiments, p < 0.05
1.0
0.8
0.6
0.4
0.2
0.0
5%
3 vs 3
5 vs 5
2×
3 vs 3
4×
3 vs 3
2×
5 vs 5
4×
5 vs 5
Animals per group, true fold difference (300 simulations each)
F
Power by size of difference
unique density, CV 0.95; this paper's contrasts are 38–141-fold
Power at α = 0.05
1.0
0.8
0.6
0.4
0.2
0.0
2-fold
5-fold
10-fold
50-fold
3
5
8
10
15
20
30
40
60
Animals per group